

VLSI Design Internship – Task 2

Topic: RTL Design of Basic Combinational Circuits

1. Basic Logic Gates

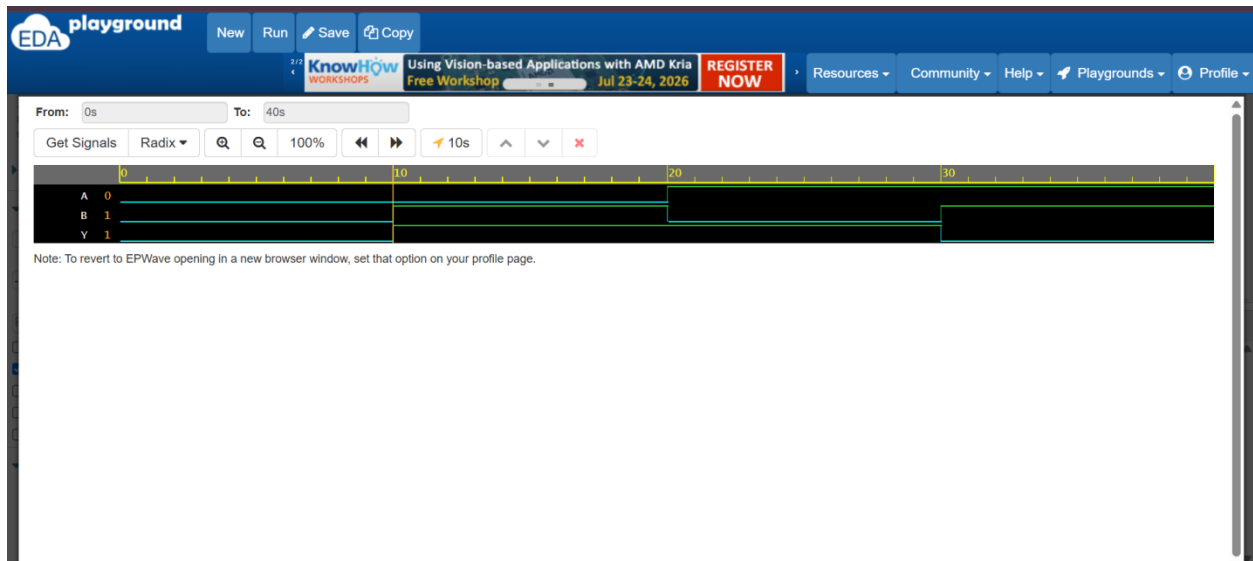
Design Logic:

The fundamental digital logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) were designed using Dataflow modeling in Verilog. Continuous `assign` statements were utilized alongside standard bitwise and logical operators to construct the RTL models. Each gate was instantiated as a separate module to ensure modularity and clean schematic-level abstraction.

Simulation Results:

A unified testbench was created to drive all 2-bit input combinations (00, 01, 10, 11) into the gate modules.

- The EPWave waveforms confirm that the outputs perfectly match standard Boolean truth tables.
- For instance, the AND gate output correctly asserts a 1 only when both inputs are 1, while the XOR gate output asserts a 1 only when the inputs are different.



2. Half Adder

Design Logic:

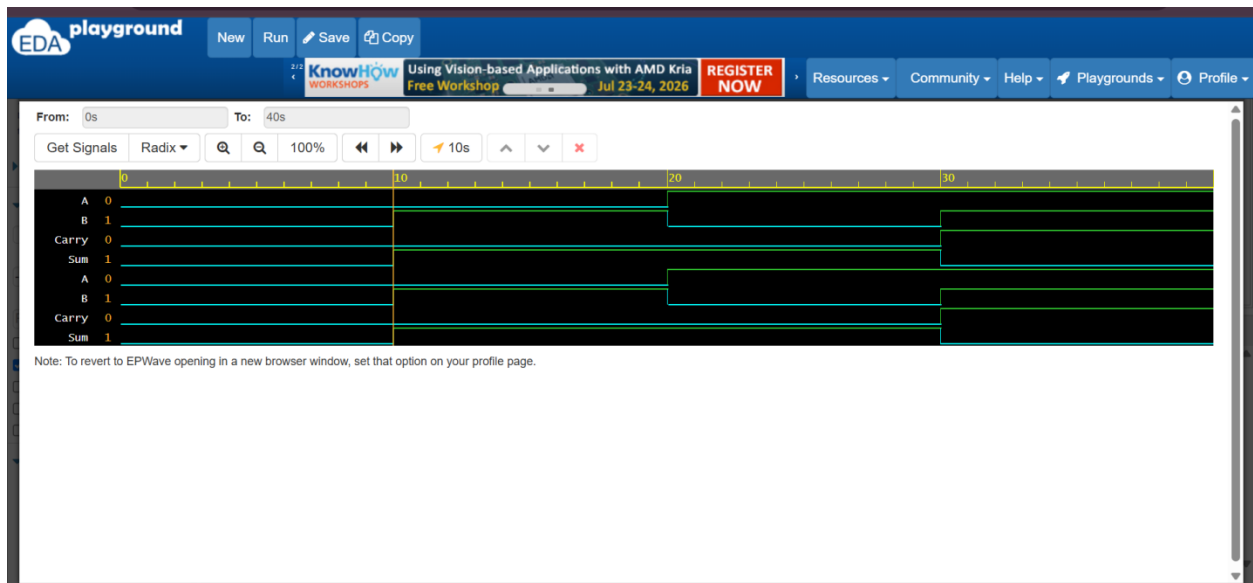
A Half Adder circuit is designed to add two single-bit binary numbers. It was modeled in Verilog using dataflow abstraction with two continuous assignments based on the following Boolean equations:

- **Sum:** Generated using an XOR operation ($\text{Sum} = A \oplus B$)
- **Carry:** Generated using an AND operation ($\text{Carry} = A \cdot B$)

Simulation Results:

The testbench simulated the four possible input combinations. The waveform results successfully verify the Half Adder truth table:

- When Inputs $A=0, B=1$ or $A=1, B=0$, the output yields $\text{Sum} = 1$ and $\text{Carry} = 0$.
- When Inputs $A=1, B=1$, the output correctly shifts to $\text{Sum} = 0$ and $\text{Carry} = 1$.



3. Full Adder

Design Logic:

A Full Adder circuit expands upon the Half Adder by incorporating a Carry-in bit, allowing for the addition of three single-bit numbers (A , B , and Cin). The RTL model was implemented using the following Boolean logic equations:

- **Sum:** $\text{Sum} = A \oplus B \oplus Cin$
- **Carry-Out:** $\text{Cout} = (A \cdot B) + (B \cdot Cin) + (A \cdot Cin)$

Simulation Results:

The testbench generated all 8 possible test vectors (from 000 to 111) applying a 10ns time delay between each state. The simulated waveforms accurately reflect the Full Adder behavior. Notably, when all three inputs are high (A=1, B=1, Cin=1), the waveforms correctly display both outputs asserting high (Sum=1, Cout=1).

